

IEEE STD 830-1998 / ISO/IEC/IEEE 29148:2018 STANDARD

Software Requirements Specification (SRS)

Crypto Crime Intelligence Tool (CCIT)

Autonomous Financial Crime Analysis, Entity Clustering, and Money-Laundering Typology Detection

Executive Abstract: Criminal networks and illicit actors increasingly leverage pseudonymous blockchain ledgers to obfuscate the proceeds of cyber extortion, ransomware, darknet trafficking, and sanctions evasion. The **Crypto Crime Intelligence Tool (CCIT)** provides law enforcement investigators with an automated, explainable analytical pipeline to trace multi-hop asset flows, cluster related wallets into real-world entities, and detect complex laundering typologies (including peel chains, rapid pass-through layering, and smurfing/fan-out structuring). CCIT operationalizes Section 4 of the national Police-AI architectural framework by unifying public blockchain data ingestion, graph intelligence, and strict **Human-In-The-Loop (HITL)** review into a court-admissible decision-support system.

SYSTEM CODENAME

CCIT / Project Cybersleuth

INTEGRATED POLICE-AI USE CASES

UC-81, UC-80, and UC-83 (Section 4 Pipeline)

AUTHOR / ENGINEERING TEAM

AI & Cybercrime Intelligence Group

DOCUMENT VERSION

1.0.0 (Submission Baseline)

EVIDENTIARY STANDARD

ISO/IEC 27037 (Digital Forensics Integrity)

TARGET BLOCKCHAIN ECOSYSTEMS

Bitcoin (UTXO), Ethereum/EVM, and Tron (TRC-20)

Governance & Compliance Statement: This tool operates exclusively on public distributed ledger data. It generates investigative leads with transparent feature-level explanations and confidence metrics; it never outputs automated legal verdicts of guilt, ensuring full adherence to LEA ethical mandates and high-risk AI accountability guidelines.

1 Framework Alignment & Pipeline Architecture

1.1 Operational Challenge & Scope

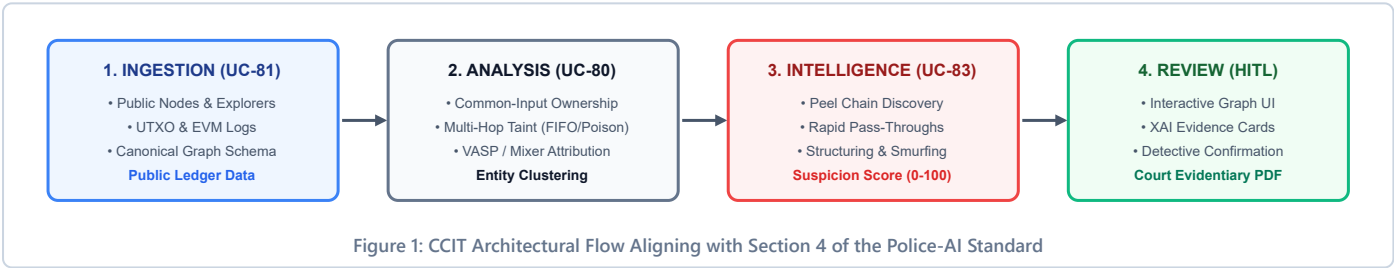
Cryptocurrency transactions are recorded on immutable public ledgers, but tracing illicit capital manually across hundreds of hops, mixers, and intermediary wallets is prohibitively labor-intensive for detectives. The CCIT automates graph synthesis, entity clustering, and anomaly detection to deliver actionable leads while preserving investigator sovereignty and legal chain of custody.

1.2 Synthesis of Three National Police-AI Framework Use Cases

Rather than delivering isolated point utilities, CCIT integrates three designated capabilities from the Police-AI reference standard into a continuous, sequential analytical pipeline:

ID	Framework Use Case	Pipeline Role	Functional Scope & Responsibilities
UC-81	Blockchain Transaction Analysis	FOUNDATION LAYER	Ingests raw multi-chain ledger blocks via public RPCs; extracts UTXO inputs/outputs and EVM smart contract event logs; normalizes records into canonical value-transfer graphs.
UC-80	Cryptocurrency Investigation	ANALYSIS LAYER	Resolves pseudonymous addresses into actor clusters using Common-Input Ownership and deposit sweep heuristics; runs multi-hop taint tracing (FIFO, Proportional, Poison); links known VASP tags.
UC-83	Money-Laundering Pattern Detection	INTELLIGENCE LAYER	Scans graph topologies for known laundering motifs (peel chains, rapid pass-through layering, fan-out smurfing); computes normalized suspicion scores (0–100) with XAI evidence cards.

1.3 End-to-End Pipeline Narrative Diagram



1.4 Foundational Governance & Ethical Principles

1. Public Ledger Exclusivity

The system relies solely on public ledger broadcasts and verified OSINT attributions. No warrant or privileged API access is required for ingestion, ensuring zero operational delays.

2. Human Primacy & Non-Automated Guilt

Under EU AI Act and LEA policy standards, CCIT is strictly advisory. An alert is never an accusation or verdict; it is an evidentiary lead requiring human detective verification.

2 Specific Functional Requirements

2.1 Foundation Layer: Blockchain Transaction Analysis (UC-81)

- FR-81.1 Multi-Chain Ingestion Engine:** Ingests blocks, transactions, and mempools from Bitcoin (UTXO), Ethereum/EVM (Account model, internal smart contract calls, ERC-20 transfers), and Tron (TRC-20 transfers).
- FR-81.2 Data Normalization Pipeline:** Normalizes heterogeneous ledger entries into a unified schema: tx_hash, network, source_address, destination_address, asset_symbol, raw_amount, fiat_usd_value (block-time historical rate), timestamp, and block_height.
- FR-81.3 Dynamic Graph Construction:** Synthesizes directed multigraphs $G = (V, E)$ from a target seed address A_0 up to K hops ($1 \leq K \leq 10$), filtering by asset type and execution date range.

2.2 Analysis Layer: Cryptocurrency Investigation & Entity Clustering (UC-80)

- FR-80.1 UTXO Common-Input Clustering:** Employs Disjoint-Set Union (Union-Find) algorithms to group all input addresses in a multi-input Bitcoin transaction into a single unified EntityCluster based on cryptographic co-signing proof.
- FR-80.2 EVM Sweep & Pass-Through Heuristic:** Identifies and clusters intermediate EVM addresses with < 5 lifetime transactions that forward > 98% of received funds within 5 blocks of receipt.
- FR-80.3 Multi-Hop Taint Propagation:** Supports three selectable taint models: *FIFO (First-In, First-Out)* for chronological fund routing; *Proportional/Haircut* for linear dilution tracking; and *Poison Taint* for total contamination tracking.
- FR-80.4 Attribution Tagging:** Cross-references all graph nodes against known databases (OFAC sanctions, licensed VASPs/exchanges, privacy mixers, and bridge routers).

2.3 Intelligence Layer: Money-Laundering Pattern Detection (UC-83)

The intelligence engine evaluates active subgraphs against four established financial crime topologies:

Typology	Mathematical Condition & Graph Signature	Severity	Scoring Weight
Peel Chain	Directed path of length $L \geq 5$ where each hop forwards > 90% of balance to a fresh address while "peeling" < 10% to an unverified secondary address.	HIGH	$w = 0.35$
Rapid Pass-Through (Layering)	Intermediary wallet where holding time $\Delta t < 600$ seconds and residual balance is < \$10 USD, indicating zero intention of sustained custody.	HIGH	$w = 0.30$
Smurfing / Structuring	Bipartite graph motif: 1 source address fans out to N intermediate wallets ($N \geq 4$), which fan in to a common aggregator within ≤ 48 hours, conserving $\geq 90\%$ volume.	CRITICAL	$w = 0.40$
Mixer & Bridge Hop	Direct transaction with anonymization protocols (e.g., Tornado Cash) or unmonitored cross-chain bridges without clear onwards destination tracking.	CRITICAL	$w = 0.50$

- FR-83.2 Calibrated Suspicion Scoring:** Assigns a normalized score $S \in [0, 100]$ to each cluster. Scores ≥ 60 trigger high-priority investigative alerts for detective review.
- FR-83.3 Explainable AI (XAI) Evidence Cards:** Translates mathematical graph anomalies into structured natural-language rationale explaining velocity, retention ratios, and transaction counts.

2.4 Presentation & Human-In-The-Loop Verification

- FR-4.1 Interactive Force-Directed Canvas:** Renders color-coded wallet clusters with real-time zooming, node expansion, and edge filtering.
- FR-4.2 Temporal Playback Scrubber:** Allows the investigator to replay fund transfers chronologically across historical blocks.
- FR-4.3 Detective Review & Annotation:** Mandates human sign-off to confirm leads, dismiss false positives with recorded rationale, and attach case warrant references.
- FR-4.4 Tamper-Evident Court PDF Export:** Generates ISO 27037-compliant forensic intelligence reports sealed with cryptographic SHA-256 digests.

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Non-Functional Requirements & Security

3.1 Performance & Scalability Targets

Metric	Target Threshold	Validation Protocol
Graph Extraction Latency	< 1.8 seconds	Retrieve 3-hop neighborhood ($\leq 2,500$ transactions) from a target seed wallet.
Typology Scan Latency	< 500 milliseconds	Execute all four rule engines and scoring equations across the active subgraph.
Canvas Rendering Capacity	$\geq 5,000$ nodes / edges	Maintain ≥ 45 FPS during interactive zooming and panning via WebGL.
Ingestion Throughput	$\geq 1,000$ tx/sec	Batch processing and normalization rate for historical block synchronization.

3.2 Security Architecture & Forensic Integrity (ISO 27037)

- NFR-SEC-1 Cryptographic Chained Audit Logging:** Every investigator action, query parameter, and review decision is stored in an append-only log where each record includes the SHA-256 hash of the preceding entry:

Hash_n = SHA-256(Hash_{n-1} || Timestamp || BadgeID || Action || Payload)

- Any retroactive database tampering breaks the hash chain, immediately exposing unauthorized alterations.
- NFR-SEC-2 Data Encryption:** All data in transit is encrypted using TLS 1.3. Sensitive case annotations, suspect cross-references, and investigator credentials are encrypted at rest with AES-256-GCM.
 - NFR-SEC-3 Role-Based Access Control (RBAC):** Access is partitioned into four strict tiers: Analyst (triage only), Lead Detective (case annotations & warrants), Legal Auditor (evidentiary sign-off), and Administrator.

3.3 Responsible AI & Legal Governance Standards

Adherence to High-Risk AI Law Enforcement Standards

1. **Absolute Human Oversight:** CCIT is strictly a decision-support utility. The software contains no automated hooks to freeze exchange accounts, issue warrants, or generate formal indictments.

2. **Algorithmic Explainability & Defensibility:** The system strictly rejects "black-box" predictions. Defense counsel and judges can audit every weight, transaction timestamp, and heuristic used in the suspicion score calculation.

3.4 Canonical System Data Schema

Entity	Primary Attributes	Role in Forensic Graph
WalletNode	address, blockchain, cluster_id, balance_usd, label	Vertex in the directed transaction graph representing a public key or smart contract.
TransferEdge	tx_hash, from_addr, to_addr, usd_val, timestamp	Directed edge representing value transfer with immutable block timestamp and fee.
EntityCluster	cluster_id, heuristic_type, member_wallets, net_flow	Aggregated group of addresses proven to be controlled by the same real-world entity.
TypologyAlert	alert_id, typology, suspicion_score, xai_summary	Forensic lead object submitted to the detective for verification and court filing.

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Verification Matrix & Formal Sign-Off

4.1 Acceptance Criteria & Verification Test Protocol

ID	Requirement Target	Verification Method	Pass / Acceptance Criteria
VR-01	Multi-Chain Ingestion (UC-81)	Automated Integration Test	100% address, balance, and transaction fee parity with live blockchain blocks.
VR-02	Common-Input Clustering (UC-80)	Historical Benchmark Dataset	Multi-input UTXO transactions resolved into single cluster ID with 100% precision.
VR-03	Typology Detection (UC-83)	Synthetic Testbed Vectors	> 95% recall on injected peel chains, rapid pass-throughs, and structuring motifs.
VR-04	Explainability (XAI)	Detective Blind Review Test	Certified financial crime analysts accurately diagnose flag cause within 30 seconds.
VR-05	Forensic Audit Export	Cryptographic Hash Verification	Generated PDF dossier SHA-256 matches chained audit log; zero verification discrepancies.

4.2 Institutional Submission & Approval Certification

By signing below, the project technical leadership and the evaluation authority verify that this Software Requirements Specification accurately formalizes the operational, functional, and ethical mandates for the **Crypto Crime Intelligence Tool (CCIT)**.

TECHNICAL LEAD / AI SYSTEMS ARCHITECT

Project Cybersleuth Engineering Team

Signature: _____

Date: _____

SUPERVISORY DETECTIVE / EVALUATION CHAIR

Financial Crime & Cyber Intelligence Division

Signature: _____

Date: _____